최적제어과제4이윤경   2021321216이윤경   과제마감일: 21.10.30

$\boxed{4.14}$ 13년 3번 기출 교재181 노트32 4.2단원 케이스 4

Find extremal curves for $J(x)=\int_0^{t_f}\frac{\sqrt{1+\dot{x}^2}}{x}\,dt$. $x(0)=0$, $x(t_f)=$ lie on $\theta(t)=t-5$. $\dot{\theta}=1$

EL식 $\rightarrow \frac{\partial g}{\partial x}-\left(\frac{\partial g}{\partial \dot{x}}\right)'=0=-\frac{\sqrt{1+\dot{x}^2}}{x^2}-\left(\frac{1}{x}\cdot\frac{2\dot{x}}{2\sqrt{1+\dot{x}^2}}\right)'=\frac{\sqrt{1+\dot{x}^2}}{x^2}+\left(\frac{\dot{x}}{x\sqrt{1+\dot{x}^2}}\right)'=\frac{\sqrt{1+\dot{x}^2}}{x^2}+\left[x^{-1}\cdot\dot{x}\cdot(1+\dot{x}^2)^{-0.5}\right]'=\frac{\sqrt{1+\dot{x}^2}}{x^2}+$ 3단미분 필기본12쪽 ㄱㄱ.

$\Rightarrow x\ddot{x}+\dot{x}^2=-1 \Rightarrow \frac{d}{dt}(x\cdot\dot{x}) \xRightarrow{\text{적분}} x\dot{x}=-t+C_1 \xRightarrow{\text{적분}} \boxed{\frac{x^2}{2}=-\frac{t^2}{2}+C_1t+\cancel{C_2}}$, $x(0)=0=0+0+C_2$, $\therefore C_2=0$

$\boxed{\therefore x=\pm\sqrt{-t^2+10t}}$

BC식 $\rightarrow \frac{x(t_f)^2}{2}=-\frac{t_f^2}{2}+C_1t_f=\frac{(t_f-5)^2}{2}=-\frac{t_f^2}{2}+C_1t_f$

$\therefore x(t_f)=\pm\frac{5}{\sqrt{2}}$

횡단식 $\rightarrow g+\frac{\partial g}{\partial \dot{x}}(\dot{\theta}-\dot{x})=0=\frac{\sqrt{1+\dot{x}^2}}{x}+\left(\frac{\dot{x}}{x\sqrt{1+\dot{x}^2}}\right)(1-\dot{x})=\frac{(\sqrt{1+\dot{x}^2})^2}{x\sqrt{1+\dot{x}^2}}+\frac{\dot{x}}{x\sqrt{1+\dot{x}^2}}-\frac{\dot{x}^2}{x\sqrt{1+\dot{x}^2}}=\frac{1+\cancel{\dot{x}^2}+\dot{x}-\cancel{\dot{x}^2}}{x\sqrt{1+\dot{x}^2}}=1+\dot{x}$, $\therefore \dot{x}(t_f)=-1$

$t=t_f$ $\boxed{\dot{x}(t_f)=-1}$

$5+\frac{5}{\sqrt{2}}$

④ $x(t_f)=t_f-5$
$\dot{x}(t_f)=-1$

$\boxed{x^2+t^2=2C_1t}$
$2x\dot{x}+2t=2C_1$
$x\dot{x}+t=C_1=(t_f-5)\cdot(-1)+t_f=C_1=5.$

$(t_f-5)^2=-t_f^2+10t_f$, $t_f=\Big]=8.53553$ or 1.46447
$x^2-10x+12.5=0$, $x=$

$5-\frac{5}{\sqrt{2}}$

$\boxed{4.14 \text{ 문제의 } 3\text{단 미분}}$ 필기본 6쪽 ㄱㄱ. 7

$\frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)=\left(\frac{\dot{x}}{x\sqrt{1+\dot{x}^2}}\right)'=\frac{\ddot{x}}{x\sqrt{1+\dot{x}^2}}-\frac{\dot{x}^2}{x^2\sqrt{1+\dot{x}^2}}-\frac{\dot{x}^2\ddot{x}}{x(1+\dot{x}^2)^{1.5}}=\frac{\ddot{x}(1+\dot{x}^2)}{x(1+\dot{x}^2)^{1.5}}-\frac{\dot{x}^2(1+\dot{x}^2)/x}{x(1+\dot{x}^2)^{1.5}}-\frac{\dot{x}^2\ddot{x}}{x(1+\dot{x}^2)^{1.5}}=\frac{\ddot{x}+\cancel{\dot{x}^2\ddot{x}}-\cancel{\dot{x}^2\ddot{x}}-\frac{\dot{x}^2}{x}-\frac{\dot{x}^4}{x}}{x(1+\dot{x}^2)^{1.5}}$

$\therefore \frac{\partial g}{\partial x}-\frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)=-\frac{\sqrt{1+\dot{x}^2}}{x^2}-\frac{\ddot{x}-\frac{\dot{x}^2}{x}-\frac{\dot{x}^4}{x}}{x(1+\dot{x}^2)^{1.5}}=\frac{-\frac{1}{x}-\frac{\dot{x}^2}{x}-\ddot{x}}{x(1+\dot{x}^2)^{1.5}}=0$, $\therefore \boxed{x\ddot{x}+\dot{x}\dot{x}=-1}$

$\boxed{4.16}$ Find extremal for $J(x)=\int_0^{t_f}\frac{\sqrt{1+\dot{x}^2}}{x}\,dt$, $x(0)=0$
$x(t_f)=$ 원 $(t-9)^2+x^2=9$ 위에 있다.

EL식 $\rightarrow \frac{\partial g}{\partial x}-\left(\frac{\partial g}{\partial \dot{x}}\right)'=0=-\frac{\sqrt{1+\dot{x}^2}}{x^2}-\left(\frac{1}{x}\cdot\frac{2\dot{x}}{2\sqrt{1+\dot{x}^2}}\right)'=\frac{\sqrt{1+\dot{x}^2}}{x^2}+$ 3단미분. 필기본12쪽 ㄱㄱ.

$\Rightarrow x\ddot{x}+\dot{x}^2=-1 \longrightarrow \frac{d}{dt}(x\dot{x}) \xrightarrow{\text{적분}} x\dot{x}=-t+C_1 \xrightarrow{\text{적분}} \frac{x^2}{2}=-\frac{t^2}{2}+C_1t+\underbrace{C_2}_{0}$

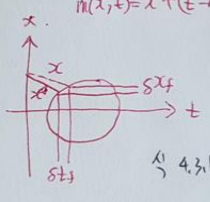
$\boxed{x^2+t^2=0=18t-72}$
$x^2+t^2-18t+72=0$

i) 수식풀이)
$m(x,t)=x^2+(t-9)^2-9=0.$

$|m(x^*,t)|_{t=t_f}=0$, $m(x^*(t_f)+\delta x_f, t_f+\delta t_f)=\cancel{x^{*2}(t_f)}+2x^*(t_f)\delta x_f+\cancel{(\delta x_f)^2}+\cancel{t_f^2}+2t_f\delta t_f-18t_f+\cancel{(\delta t_f)^2}-18\delta t_f+72$ $m(x^*(t_f),t_f)=0$, 이차계수$=0$

$\Rightarrow 2t_f\delta t_f-18\delta t_f+2x^*(t_f)\delta x_f=2x^*(t_f)\delta x_f+2(t_f-9)\delta t_f=0$ (관계식).

ii) 기하풀이)
$m(x,t)=x^2+(t-9)^2-9=0.$



x와 x^*을 그렸다. 그림으로 $\delta x_f=a\delta t_f$이다. $a=$기울기 (점 $x^*(t_f),t_f$에서).

$\cancel{2}x^*(t_f)\delta x_f+\cancel{2}(t_f-9)\delta t_f=0 \longrightarrow \boxed{\delta x_f=\frac{(9-t_f)\delta t_f}{x^*(t_f)}}$

식 4.3.18 $\rightarrow \frac{\partial g}{\partial \dot{x}}\cdot\delta x_f+\left(g-\frac{\partial g}{\partial \dot{x}}\cdot\dot{x}\right)\delta t_f=0.$

계산으로 정리후)
① $-\dot{x}(t_f)(t_f-9)+x(t_f)=0$, ② $\frac{x^2}{2}=-\frac{t^2}{2}+C_1t$, ③ $x\dot{x}=-t+C_1$, ④ $(t_f-C_1)(t_f-9)+x^2(t_f)=0.$
$\rightarrow$ ②와 ④ 식으로, $C_1t_f-9t_f+9C_1=0.$ and $18t_f-2C_1t_f-72=0.$ $\xrightarrow{\text{연립}}$ $C_1=4$, $t_f=7.2$, $x^2=-t^2+8t$, $x=\pm\sqrt{-t^2+8t}$

[3] Find the control history $u(t)$ and State history $x_1(t)$ and $x_2(t)$ to extremize the performance index:

$$J=\frac{1}{2}\int_0^1 x_1^2+x_2^2+u^2\,dt,\quad \dot{x}_1=x_2,\quad \dot{x}_2=-x_1-x_2+u$$

→ case ⓐ $x_1(0)=1,\ x_1(1)=0\ /\ x_2(0)=0,\ x_2(1)=0$

→ case ⓑ $x_1(0)=-1,\ x_1(1)=$자유$\ /\ x_2(0)=1,\ x_2(1)=0$

$\boxed{\text{case a}}$ Solve case ⓐ and plot the solution $u(t), x_1(t), x_2(t)$.

$\boxed{\text{case b}}$ Solve case ⓑ and plot the solution $u(t), x_1(t), x_2(t)$.

Use MATLAB to get easy matrix computation

$e^A = \text{expm}(A)$.

$$H = g+\Lambda^T f = \frac{1}{2}x_1^2+\frac{1}{2}x_2^2+\frac{1}{2}u^2+\Lambda_1 x_2-\Lambda_2 x_1-\Lambda_2 x_2+\Lambda_2 u$$

optimal 식 $\longrightarrow \dfrac{\partial H}{\partial u}=0=u+\Lambda_2 \longrightarrow -u=\boxed{+\Lambda_2=-x_1-x_2-\dot{x}_2}$

costate 식 $\longrightarrow \dot{\Lambda}=-\dfrac{\partial H}{\partial x}^T$

→ $\dot{\Lambda}_1=-\dfrac{\partial H}{\partial x_1}=-x_1+\Lambda_2$

→ $\dot{\Lambda}_2=\dfrac{\partial H}{\partial x_2}=-x_2-\Lambda_1+\Lambda_2$

State 식 $\longrightarrow \dot{x}=\dfrac{\partial H}{\partial \Lambda}^T=f$

→ $\dot{x}_1=x_2=f_1$

→ $\dot{x}_2=-x_1-x_2+u=f_2$

$$\begin{pmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{\Lambda}_1\\ \dot{\Lambda}_2\end{pmatrix}=\underbrace{\begin{pmatrix}0&1&0&0\\-1&-1&0&-1\\-1&0&0&1\\0&-1&-1&1\end{pmatrix}}_{A}\begin{pmatrix}x_1\\x_2\\\Lambda_1\\\Lambda_2\end{pmatrix}\xrightarrow[\text{sol.}]{}\begin{pmatrix}x_1(t)\\x_2(t)\\\Lambda_1(t)\\\Lambda_2(t)\end{pmatrix}=e^{At}\begin{pmatrix}x_1(0)\\x_2(0)\\\Lambda_1(0)\\\Lambda_2(0)\end{pmatrix},\quad \underset{t=1}{e^{At}}=\begin{pmatrix}0.5858&0.6524&0.1663&-0.4972\\-0.8187&0.4306&0.4972&-0.9833\\-0.4845&-0.8282&0.2533&1.8131\\1.1607&-0.8171&-1.6468&2.3973\end{pmatrix}$$

case a.

$$\begin{pmatrix}x_1(1)\\x_2(1)\\\Lambda_1(1)\\\Lambda_2(1)\end{pmatrix}=\begin{pmatrix}0\\0\\\Lambda_1(1)\\\Lambda_2(1)\end{pmatrix}=e^A\cdot\begin{pmatrix}1\\0\\\Lambda_1(0)\\\Lambda_2(0)\end{pmatrix}=\begin{pmatrix}\phi_{11}(1)&\phi_{12}(1)\\\phi_{21}(1)&\phi_{22}(1)\end{pmatrix}\begin{pmatrix}1\\0\\\Lambda_1(0)\\\Lambda_2(0)\end{pmatrix},$$

$$\begin{bmatrix}0\\0\end{bmatrix}=\phi_{11}(1)\cdot\begin{bmatrix}1\\0\end{bmatrix}+\phi_{12}(1)\begin{bmatrix}\Lambda_1(0)\\\Lambda_2(0)\end{bmatrix}\longrightarrow \text{아래.}$$

$$\begin{bmatrix}\Lambda_1(0)\\\Lambda_2(0)\end{bmatrix}=\phi_{12}(1)^{-1}\cdot\left(\begin{bmatrix}0\\0\end{bmatrix}-\phi_{11}(1)\begin{bmatrix}1\\0\end{bmatrix}\right).$$

$$0=0.5858\times1+0.6524\times0+0.1663\Lambda_1(0)-0.4972\Lambda_2(0)$$
$$0=-0.8187\times1+0.4306\times0+0.4972\Lambda_1(0)-0.9833\Lambda_2(0)$$

풀고 $\Longrightarrow$ $\Lambda_1(0)=11.7473$, $\Lambda_2(0)=5.1073$ ★

$$\begin{bmatrix}\Lambda_1(1)\\\Lambda_2(1)\end{bmatrix}=\phi_{21}(1)\cdot\begin{bmatrix}1\\0\end{bmatrix}+\phi_{22}(1)\cdot\begin{bmatrix}\Lambda_1(0)\\\Lambda_2(0)\end{bmatrix}=\begin{matrix}-0.4845\times1-0.8282\times0+0.2533\times11.7473+1.8131\times5.1073=11.7511=\Lambda_1\\1.1607\times1-0.8171\times0-1.6468\times11.7473+2.3973\times5.1073=-5.9410=\Lambda_2\end{matrix}$$ ★

$-5.1073=-\Lambda_2(0)=u(0)$

$5.9410=-\Lambda_2(1)=u(1)$

$\boxed{+\text{ plot}}$

case b. $x_1(0)=1$, $\underbrace{x_1(1)=\text{자유}}$, $x_2(0)=1$, $x_2(1)=0$.

$\rightarrow$ 대신에, $\Lambda_1(1) = \left.\frac{\partial h}{\partial x_1}\right|_{t=1} = 0$. ∴(행렬의 1줄 대신 3줄 사용으로 ㄱㄱ).

$\Lambda_1(1)=0$

$$\begin{pmatrix} x_1(1) \\ x_2(1) \\ \Lambda_1(1) \\ \Lambda_2(1) \end{pmatrix} = \begin{pmatrix} x_1(1) \\ 0 \\ 0 \\ \Lambda_2(1) \end{pmatrix} = e^{A} \cdot \begin{pmatrix} 1 \\ 1 \\ \Lambda_1(0) \\ \Lambda_2(0) \end{pmatrix} = \begin{bmatrix} \phi_{11}(1) & \phi_{12}(1) \\ \phi_{21}(1) & \phi_{22}(1) \\ \phi_{31}(1) & \phi_{32}(1) \end{bmatrix} \begin{pmatrix} 1 \\ 1 \\ \Lambda_1(0) \\ \Lambda_2(0) \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \phi_{21}(1) \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \phi_{22}(1) \cdot \begin{bmatrix} \Lambda_1(0) \\ \Lambda_2(0) \end{bmatrix} \longrightarrow \text{아래}$$

$$\supset \begin{bmatrix} \Lambda_1(0) \\ \Lambda_2(0) \end{bmatrix} = \phi_{22}(1)^{-1} \cdot \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix} - \phi_{21}(1) \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right)$$

$$0 = -0.8187 \times 1 + 0.4306 \times 1 + 0.4492 \times \Lambda_1(0) - 0.4533 \times \Lambda_2(0)$$

$$0 = -0.4845 \times 1 - 0.8282 \times 1 + 0.2533 \times \Lambda_1(0) + 1.8131 \times \Lambda_2(0)$$

$\Rightarrow$ 유도 $\Lambda_1(0) = 1.7334$, $\Lambda_2(0) = 0.4818$ ☆☆

$$\begin{bmatrix} \Lambda_1(1) \\ \Lambda_2(1) \end{bmatrix} = \phi_{31}(1) \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \phi_{32}(1) \cdot \begin{bmatrix} \Lambda_1(0) \\ \Lambda_2(0) \end{bmatrix} =$$

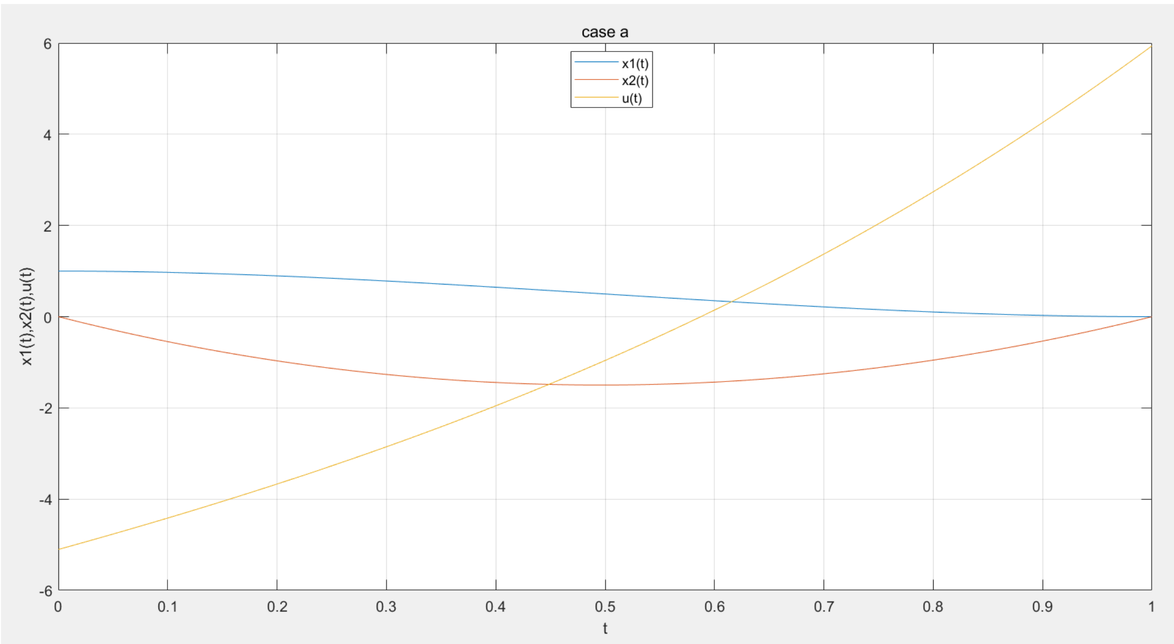
$$-0.4845 \times 1 - 0.8282 \times 1 + 0.2533 \times \underbrace{\Lambda_1(0)}_{1.7334} + 1.8131 \times \underbrace{\Lambda_2(0)}_{0.4818} = -0.0000782 = \Lambda_1(1)$$

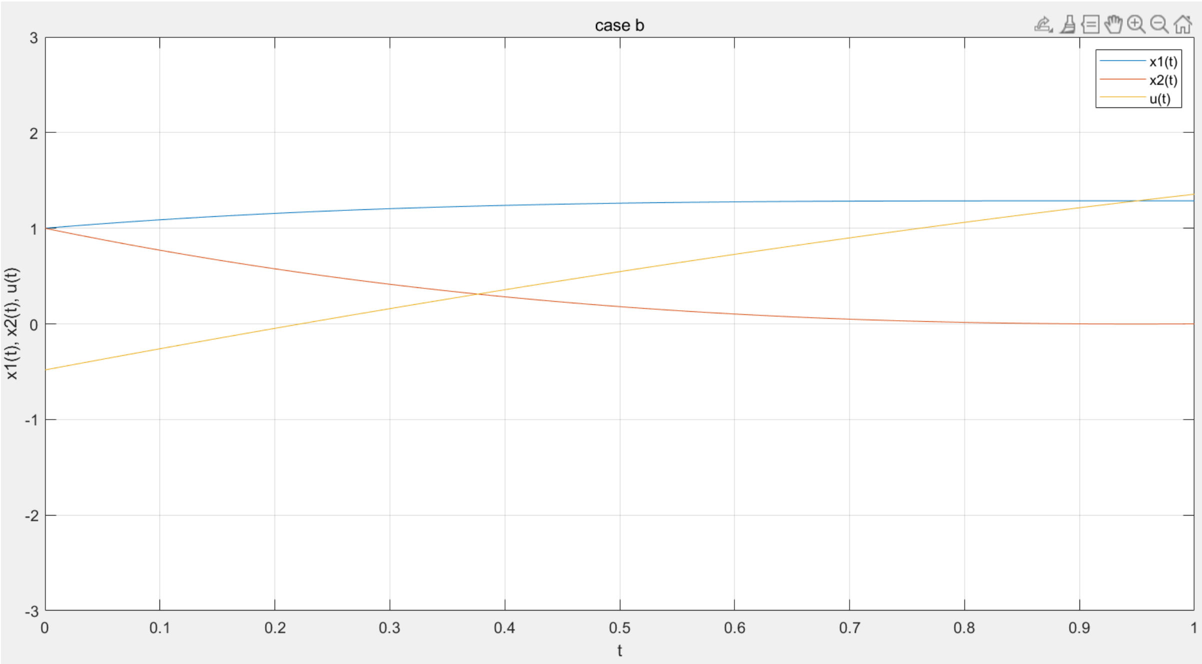
$$1.1607 \times 1 - 0.8171 \times 1 - 1.6468 \times \underbrace{\Lambda_1(0)}_{1.7334} + 2.3973 \times \underbrace{\Lambda_2(0)}_{0.4818} = -1.3559 = \Lambda_2(1)$$ ☆

$-0.4818 = -\Lambda_2(0) = u(0)$

$1.3559 = -\Lambda_2(1) = u(1)$

$\boxed{+\text{plot.}}$




case b
x1(t)
x2(t)
u(t)
x1(t), x2(t), u(t)
t
3
2
1
0
-1
-2
-3
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1